

Multipath TCP

Mukulika Maity

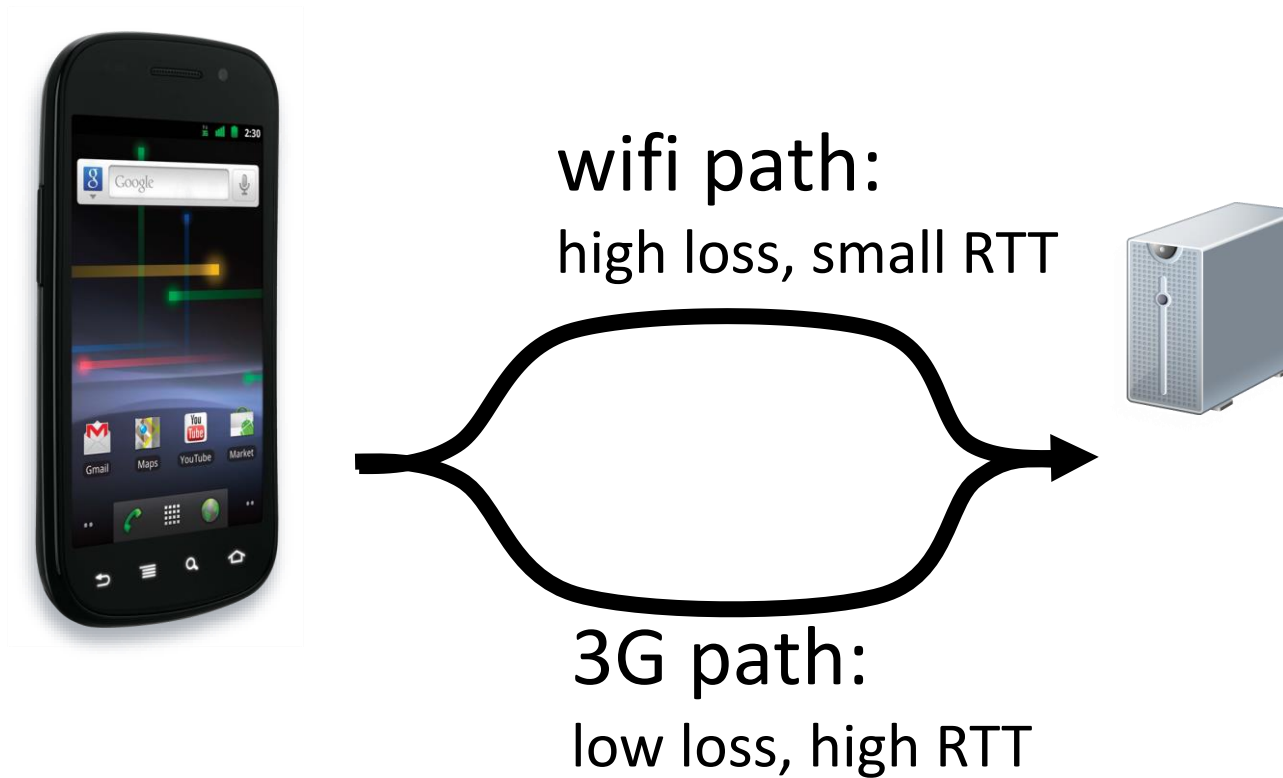
Multipath

- Mobile user
 - WiFi and cellular at the same time
- High-end servers
 - Multiple Ethernet cards
- Data centers
 - Rich topologies with many paths
- Benefits of multipath
 - Higher throughput
 - Failover from one path to another
 - Seamless mobility

Bringing Multipath to the End Host

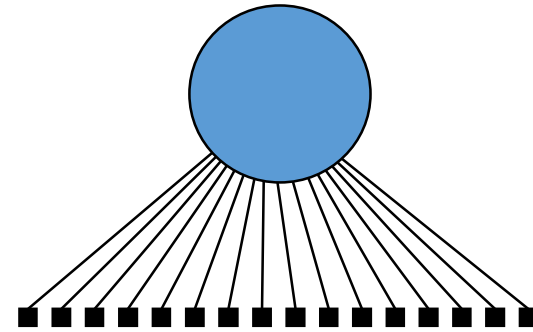
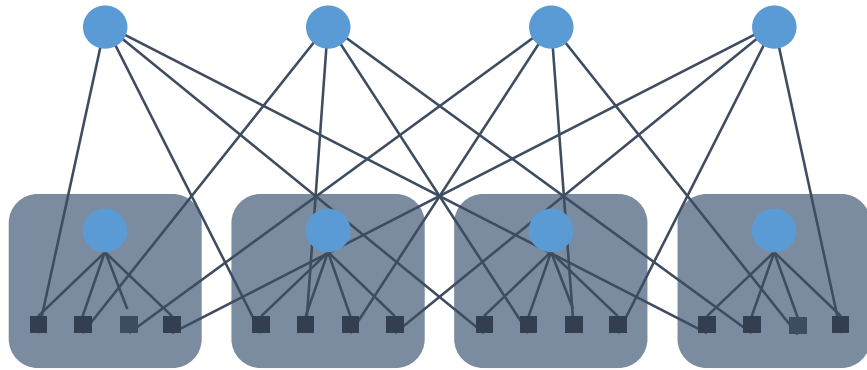
- Multiple addresses
 - One or more addresses at an end host
 - E.g., one per interface card
- Multiple paths
 - Sequence of links between sender and receiver
 - E.g., four-tuple of source and dest address and port
- Multiple subflows
 - Flow of TCP segments over an individual path
 - All associated with a single TCP connection

How should your phone balance its traffic across very different paths?



Ack: Damon Wischik

Application: Datacenters

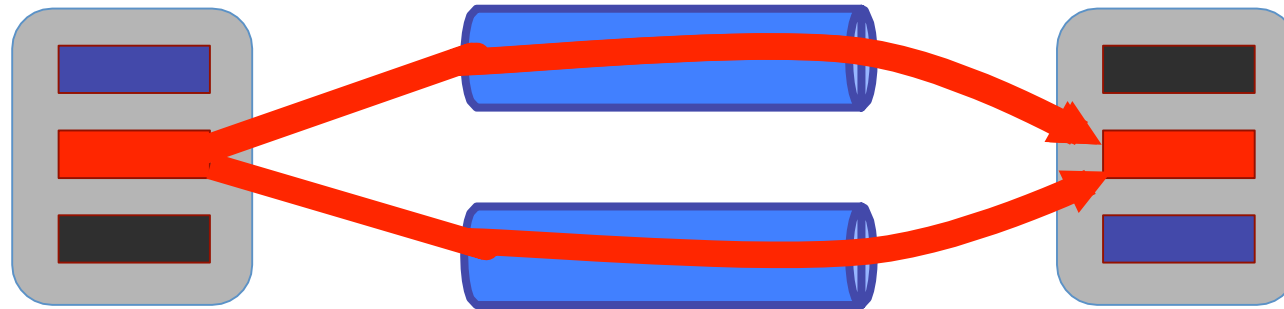


- Can we make the network behave like a large pool of capacity?

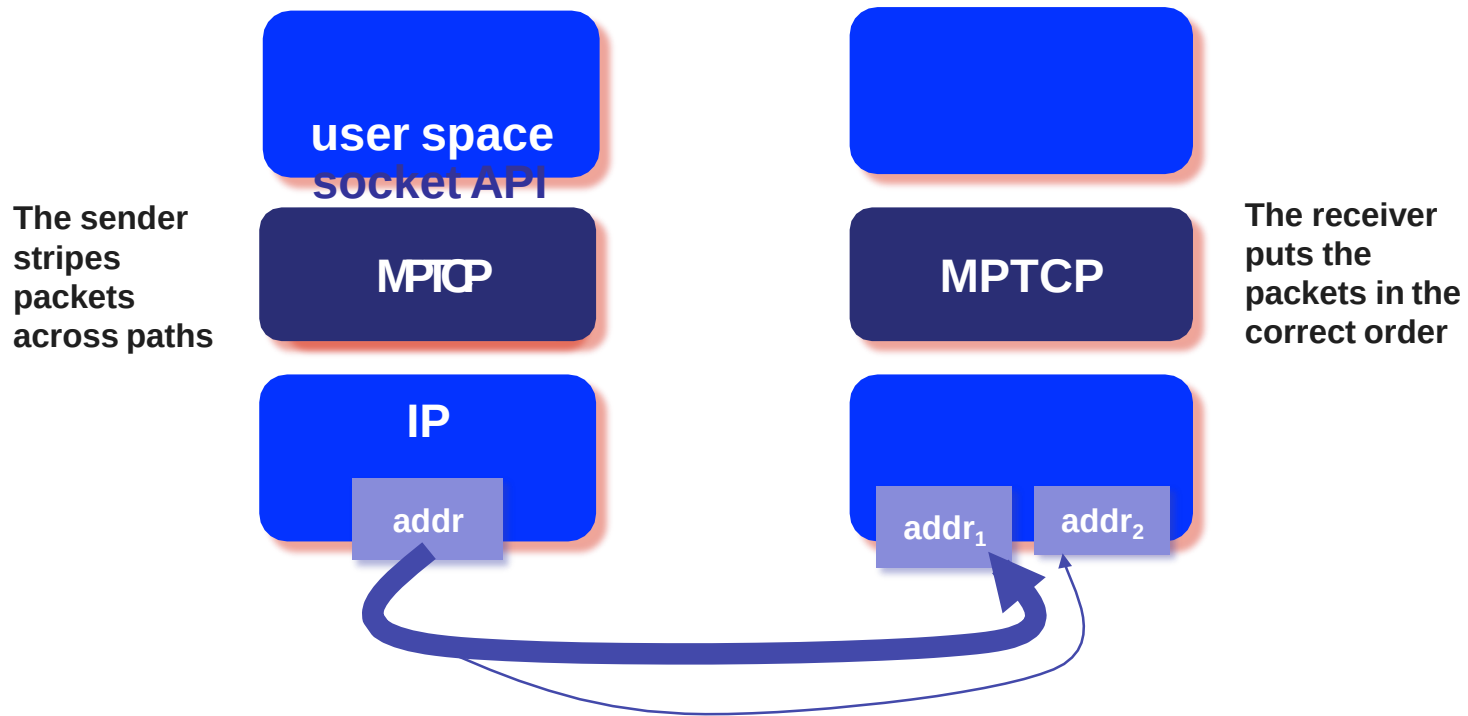
Ack: Damon Wischik

Multipath TCP

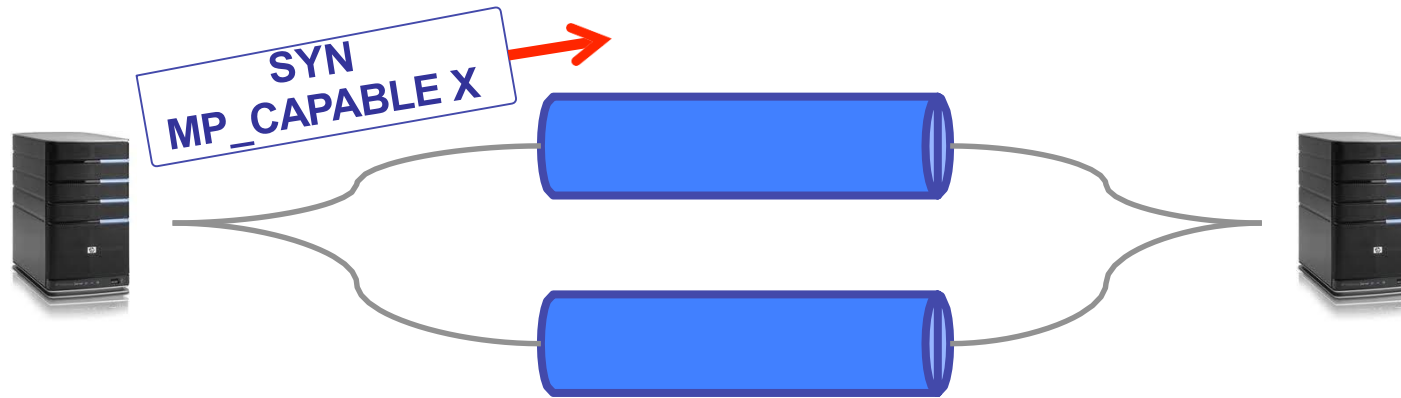
- MPTCP is a drop in replacement for TCP
 - ◆ Works with unmodified applications
 - ◆ Over the existing network



Drop-in TCP Replacement

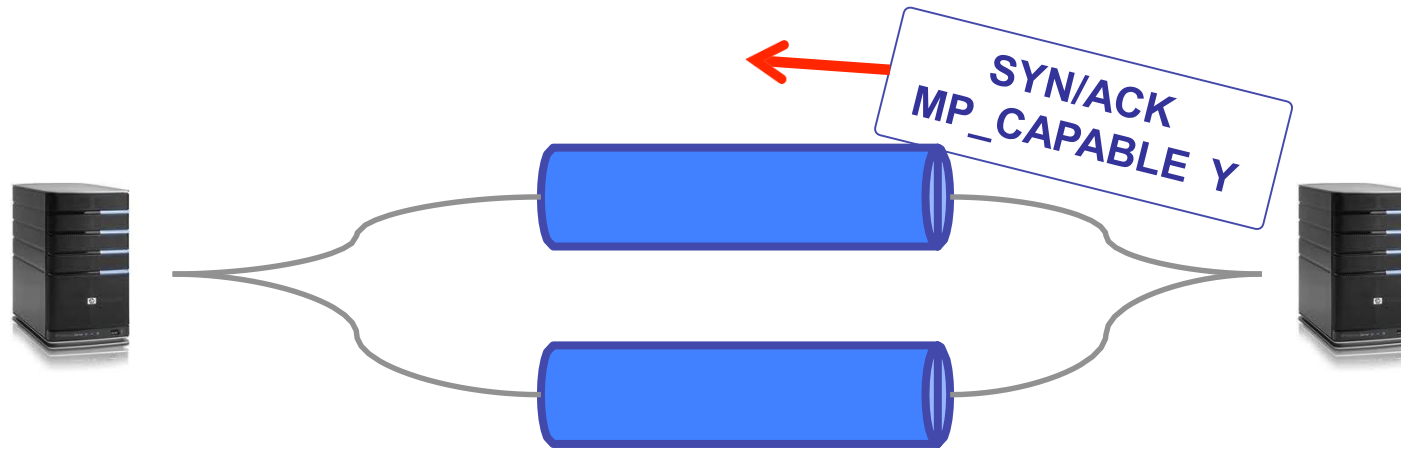


MPTCP Operation



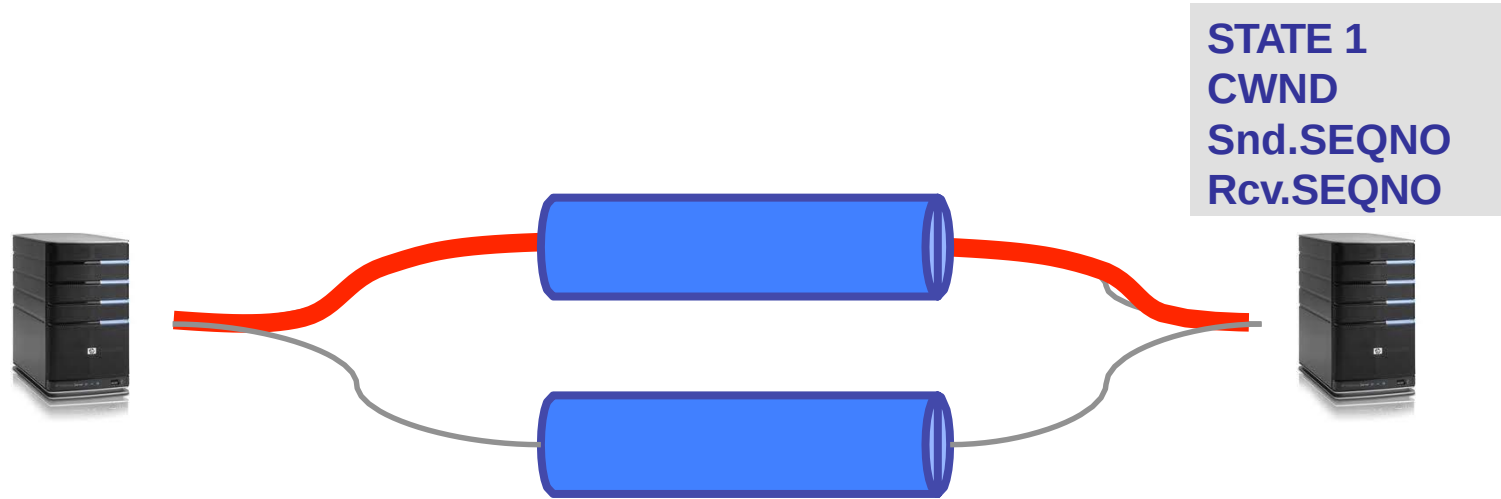
Ack: Alex C. Snoeren
CSE 222A

MPTCP Operation



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CSE 222A

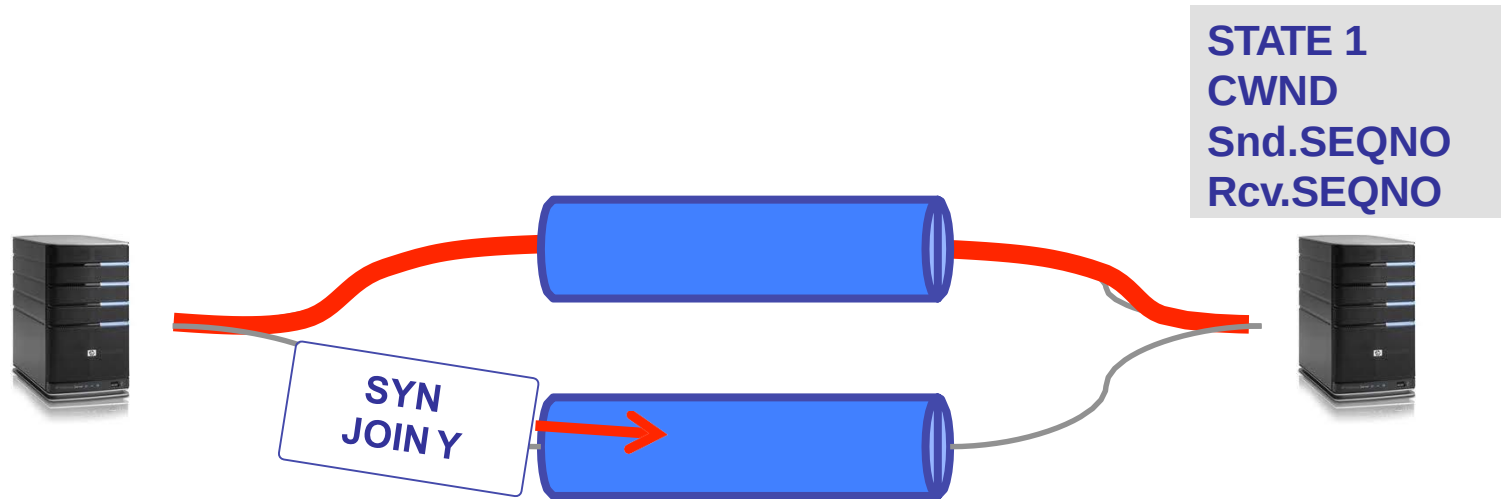
MPTCP Operation



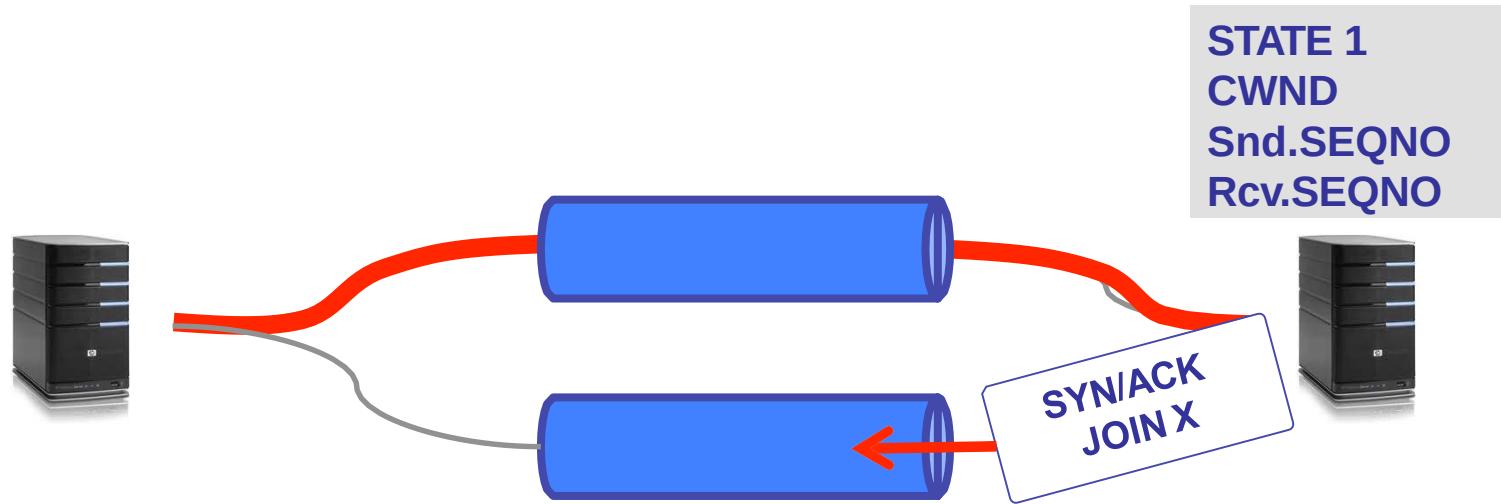
MPTCP Operation: Challenges

- Packets from second subflow use the first subflow's IP? **IP spoofing**
- Second subflow uses its own IP and start tx packets? **Firewall**
- Have 3 way handshaking
- Possible attacks? **Cryptographic key**

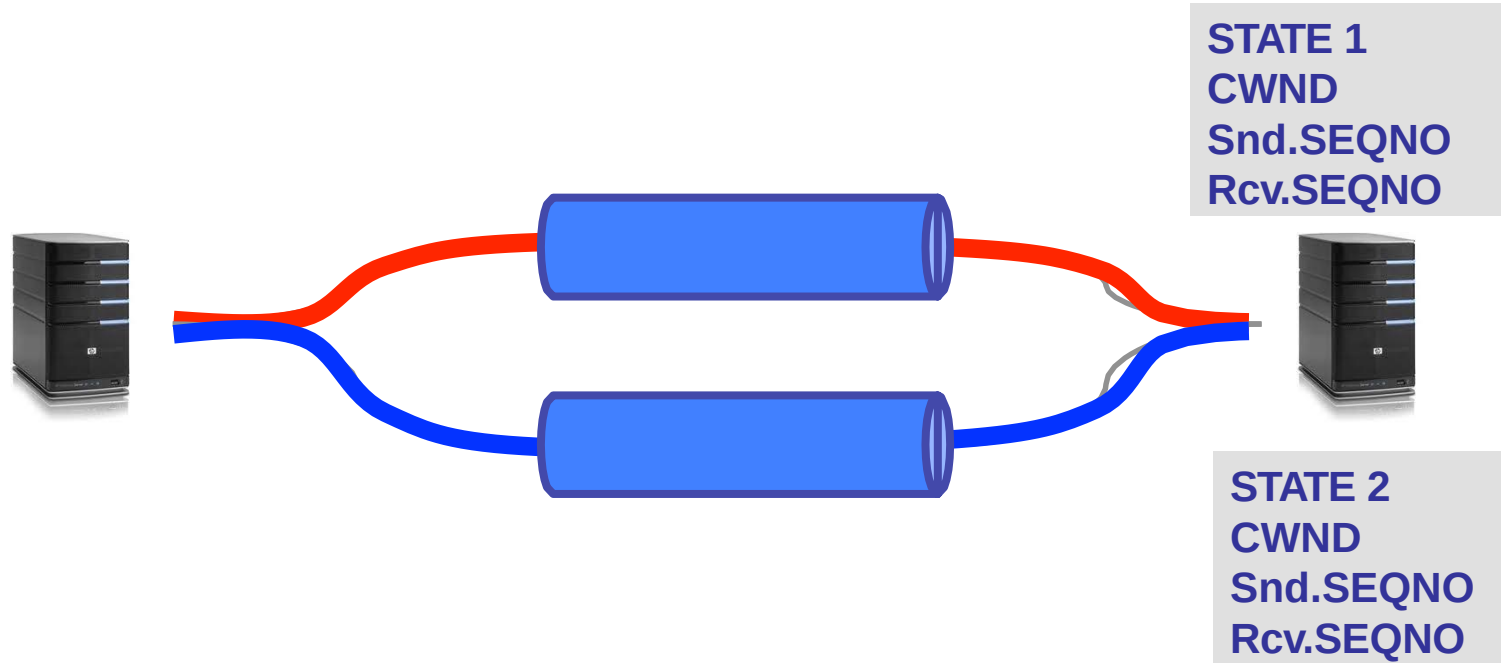
MPTCP Operation



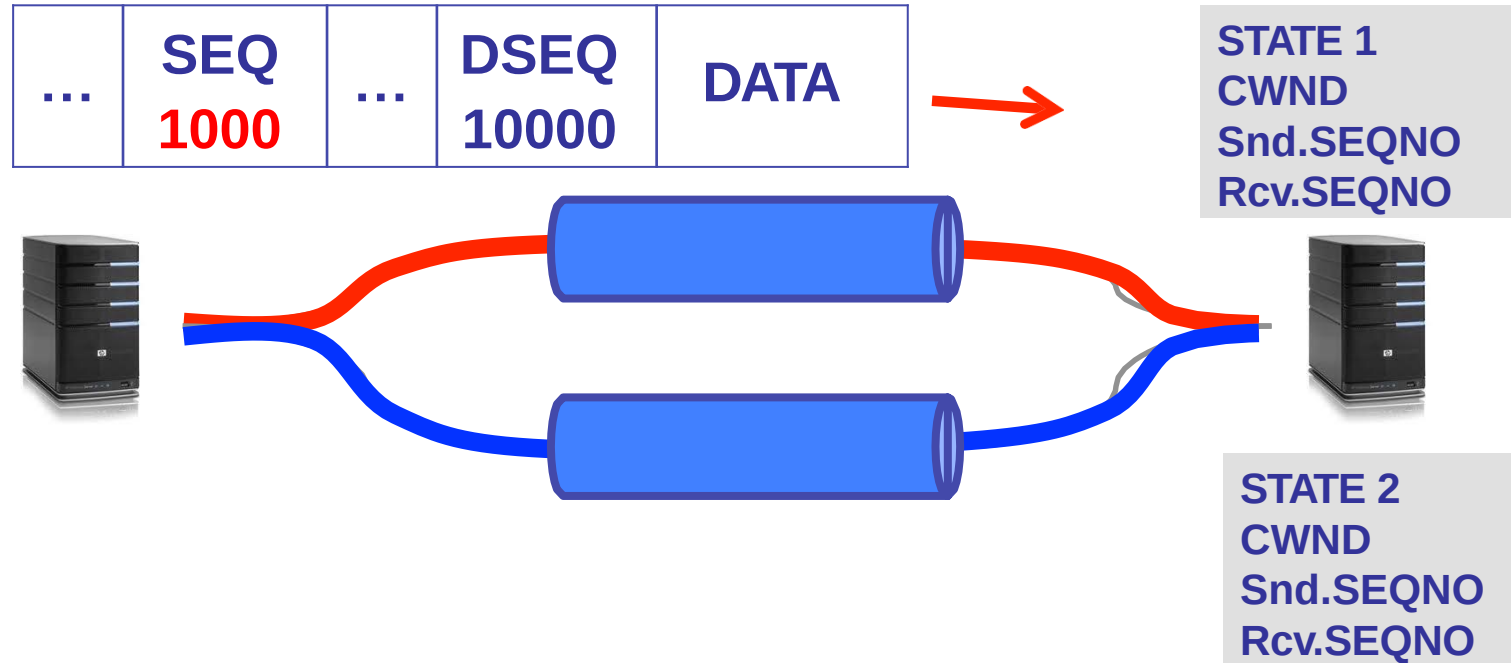
MPTCP Operation



MPTCP Operation



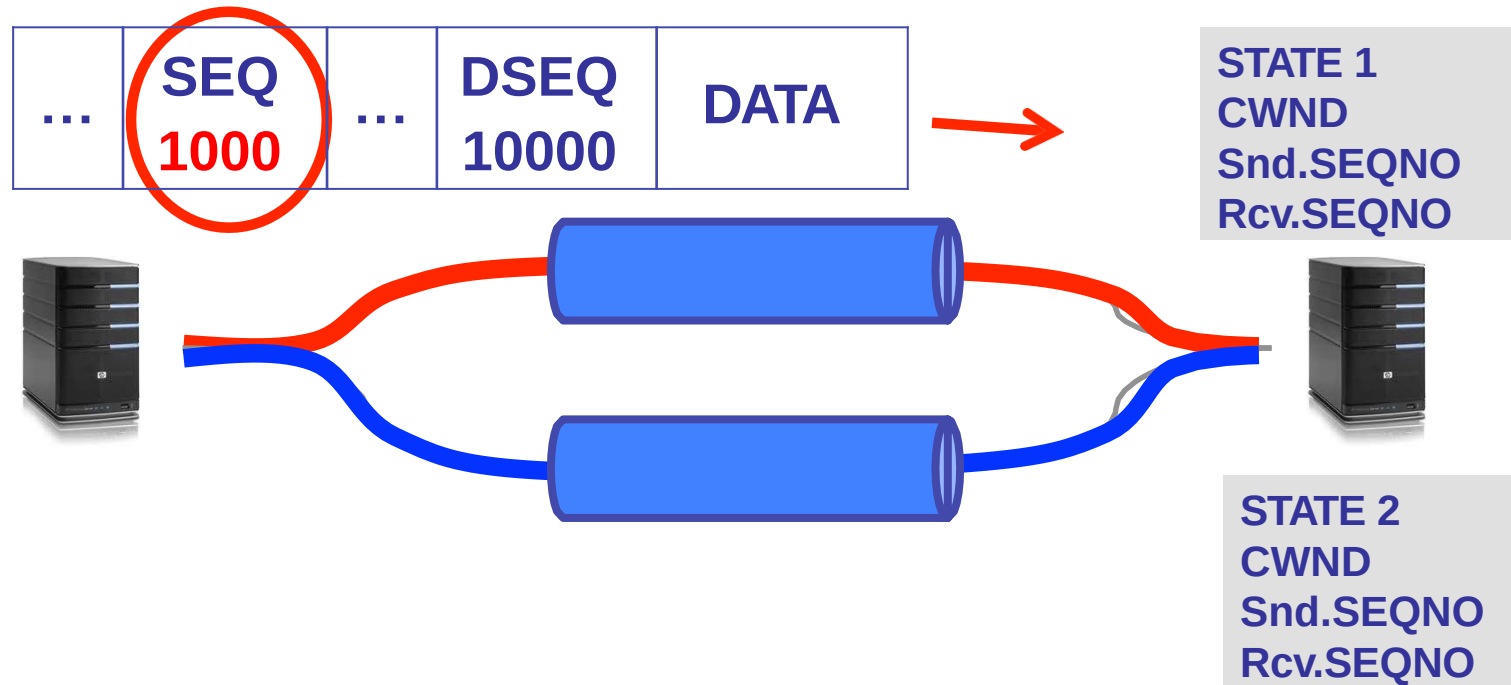
MPTCP Operation



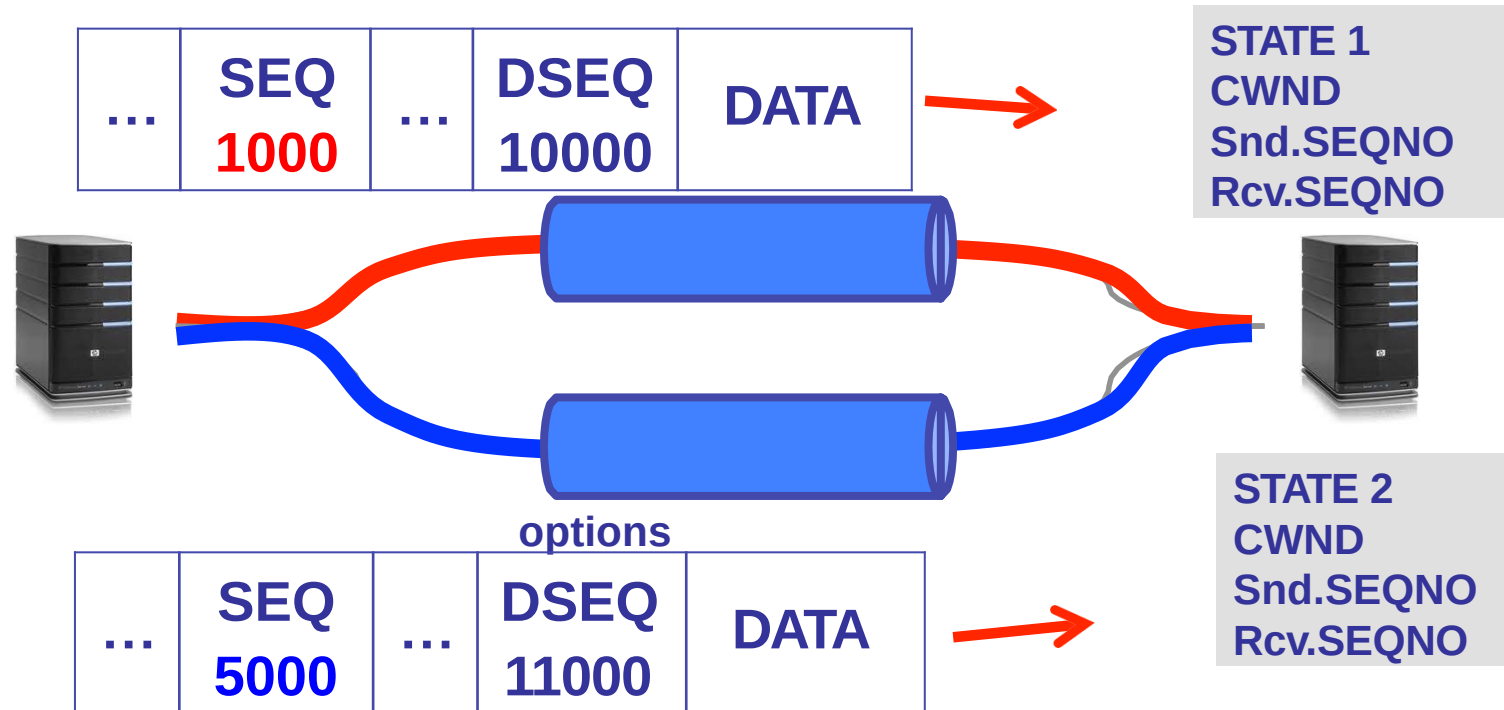
MPTCP Operation: Seq space

- Same sequence space for each subflow? **Reordering**
- Separate subflow seq space, and in TCP options data sequence number? **NIC based TSO (TCP segment offloading)**
- Mapping in the data sequence option: defining start and end of the data sequence number? **Firewall changing seq numbers**
- Mapping in the data sequence option uses relative subflow seq numbers compared to initial seq numbers instead of absolute seq numbers? **Application level gateway modifies payload and length**
- Checksum to detect that and fallback to regular TCP

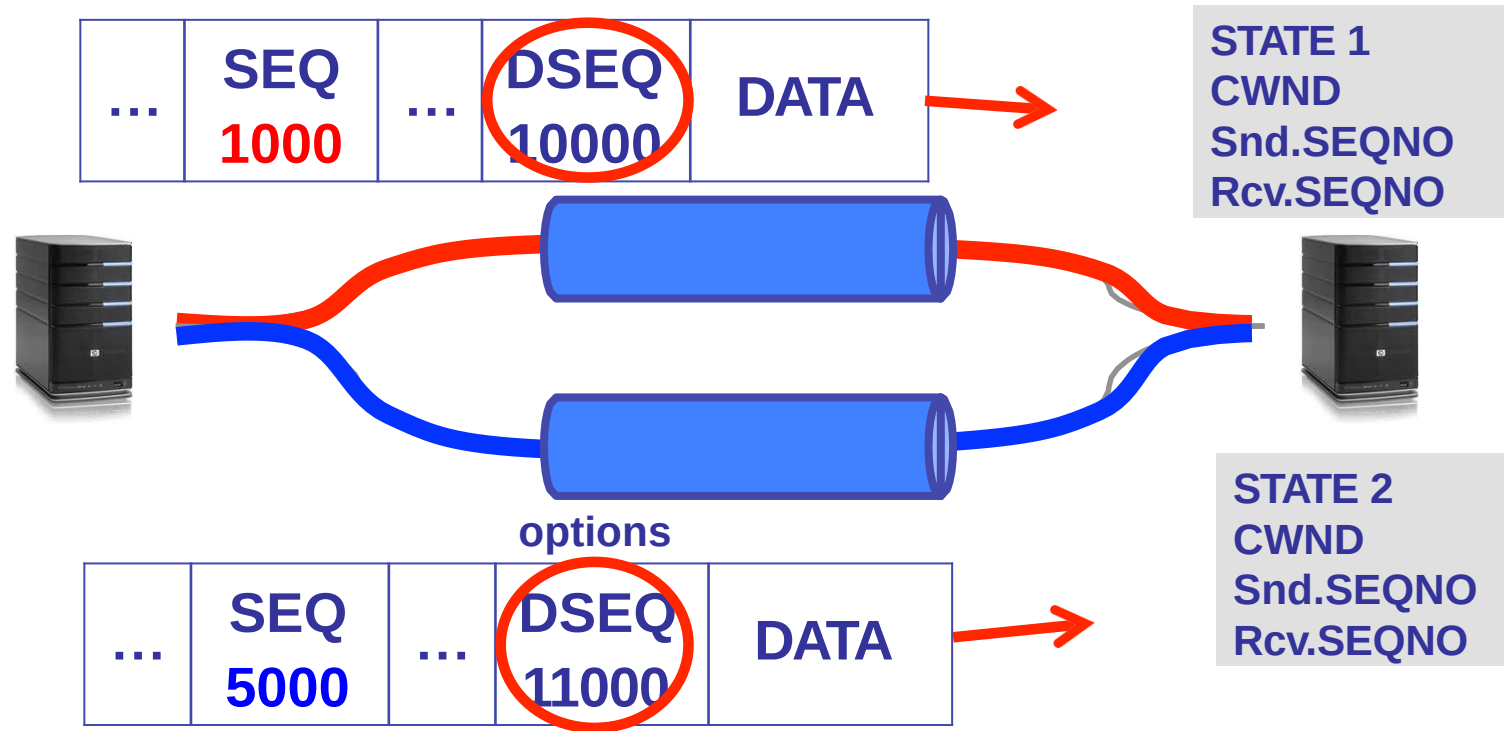
MPTCP Operation



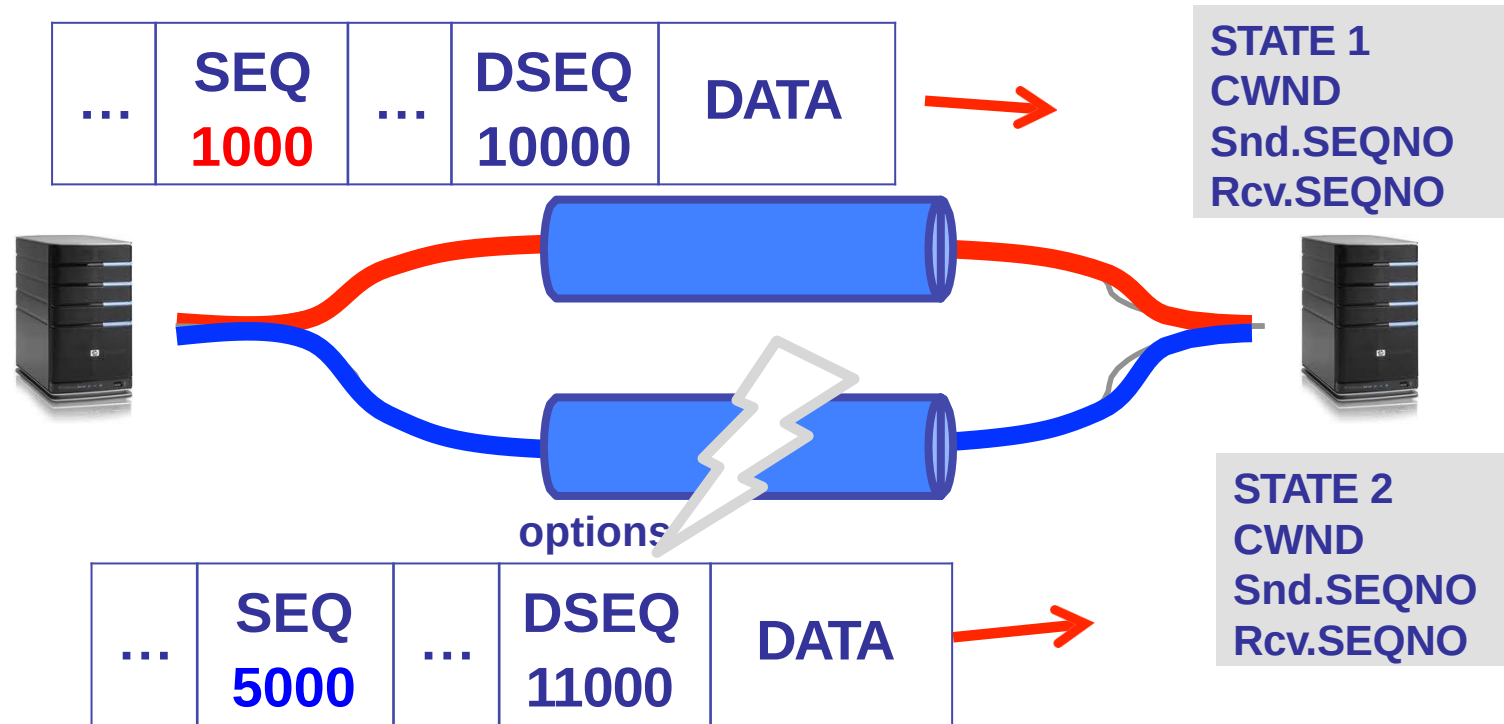
MPTCP Operation



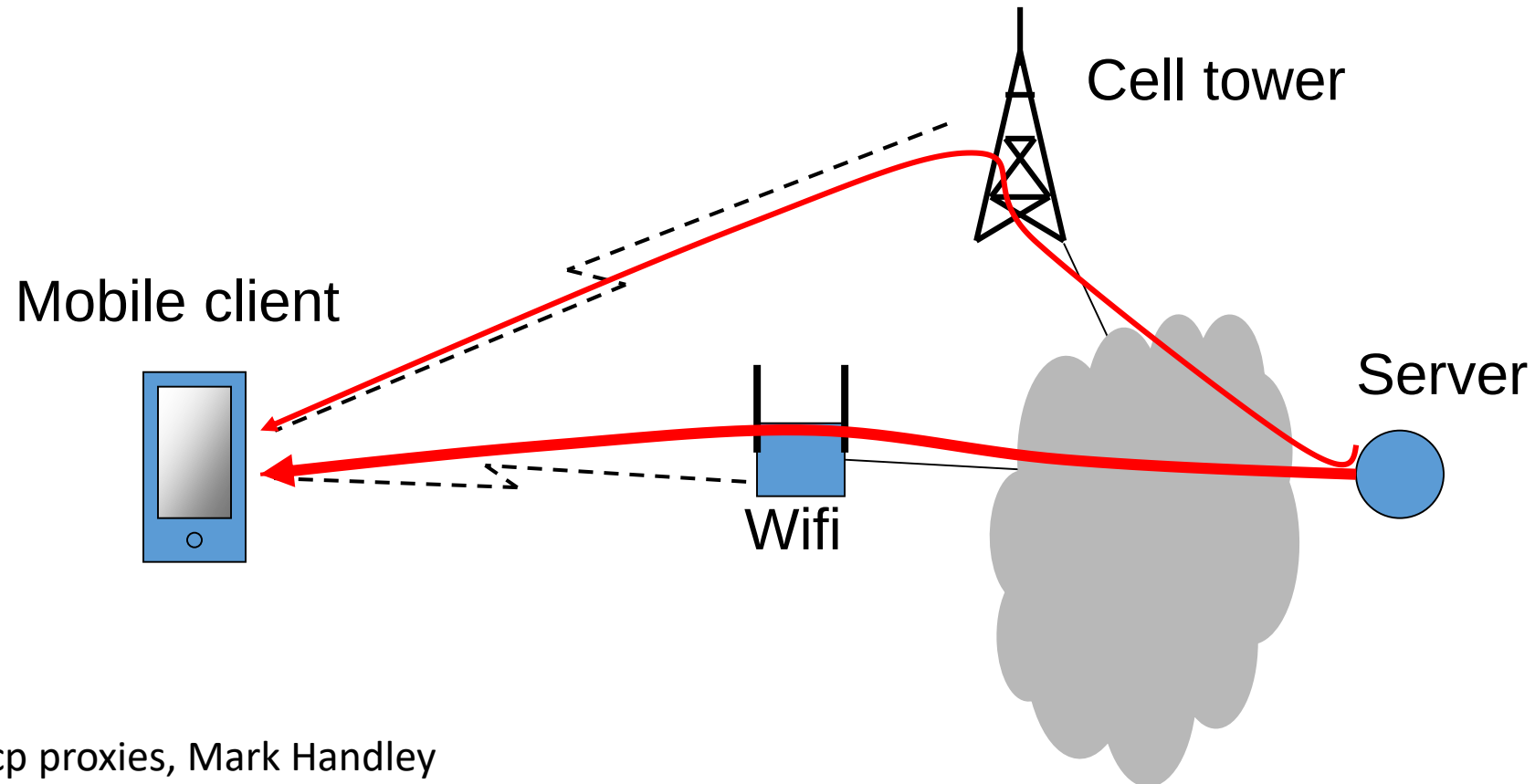
MPTCP Operation



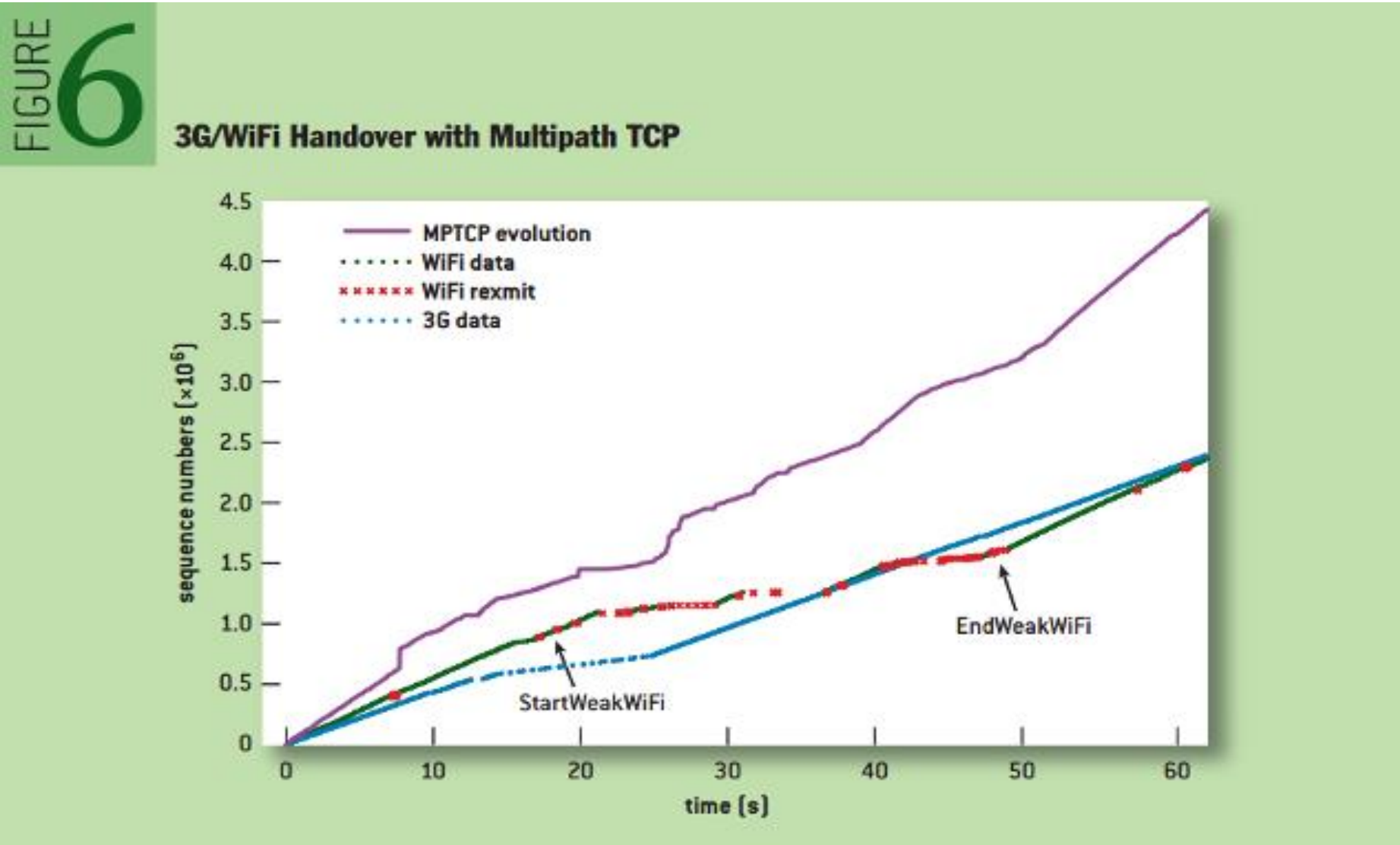
MPTCP Operation



3G/ WiFi handover with Multipath TCP



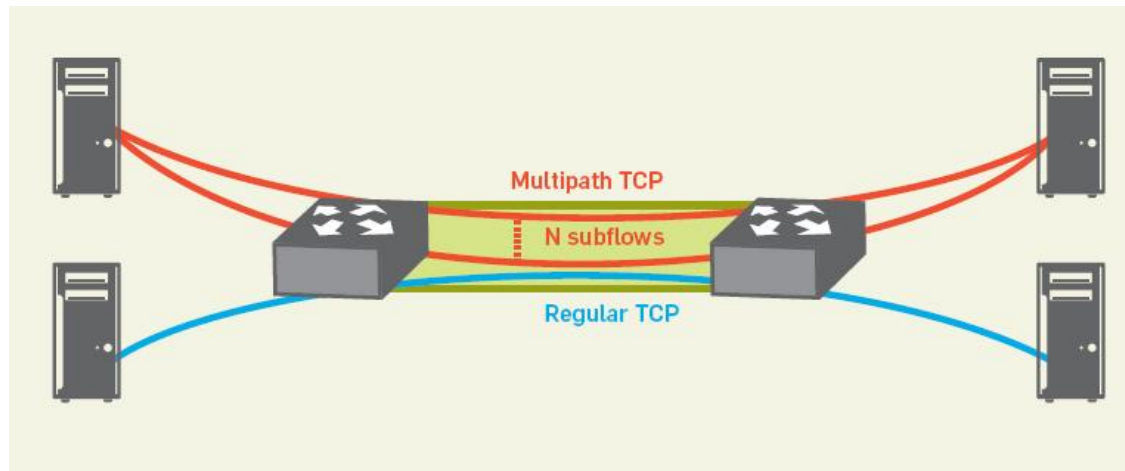
3G/ WiFi handover with Multipath TCP



Img src: Decoupled from IP, TCP is at last able to support multihomed hosts Christoph Paasch et al.

MPTCP congestion control

- MPTCP uses several paths, each path will experience different level of congestion
- Use standard TCP congestion control
- Unfairness with regular TCP



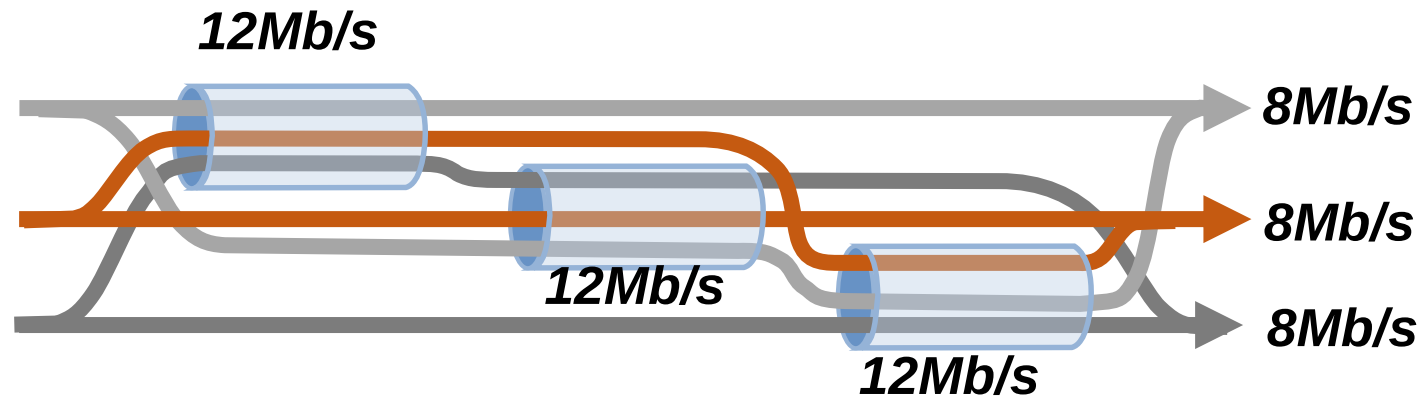
Img src:

<https://cacm.acm.org/magazines/2014/4/173230-multipath-tcp/fulltext> Christoph Paasch et al.

EWTCP congestion control

- Regular TCP
 - Each round trip, increase window w by 1
 - Each loss, decrease window w by $w/2$
- EWTCP TCP: weighted version of TCP
 - Each round trip, increase w_r by a
 - Each loss, decrease window w_r by $w_r/2$
 - w_r is the window size on path r , and $a = 1/\sqrt{n}$ where n is the number of paths

EWTCP congestion control



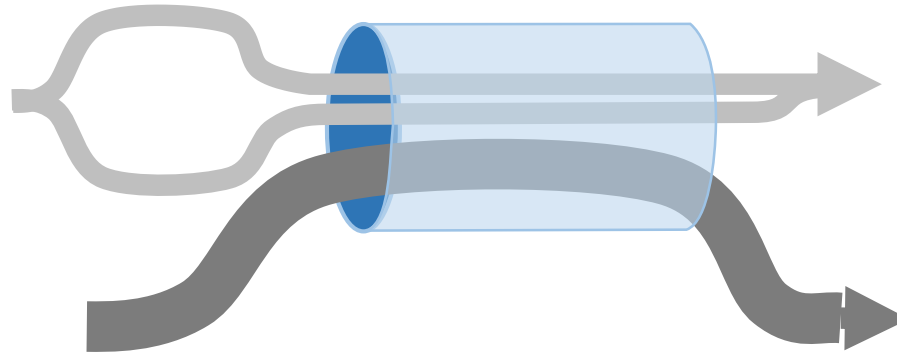
MPTCP congestion control

- [Link](#) the congestion control parameters, so as to move traffic away from the more congested paths
- Measure congestion on each subflow and move traffic away from the most congestion
- Adjust TCP congestion window on each subflow based on the level of congestion on that path.
- Congestion windows of diff subflows are coupled to ensure fairness with regular TCP

Goal #1: Fair at Shared Bottlenecks

**A multipath
TCP flow with
two subflows**

Regular TCP



To be fair, Multipath TCP should take as much capacity as TCP at a bottleneck link, no matter how many paths it is using.

MPTCP congestion control

Goal #2: Performance of all the Multipath TCP flows should be at least as that of the regular TCP: [incentive to deploy Multipath TCP](#)

Goal #2: Multipath TCP should prefer efficient paths, send more traffic on paths experiencing less congestion

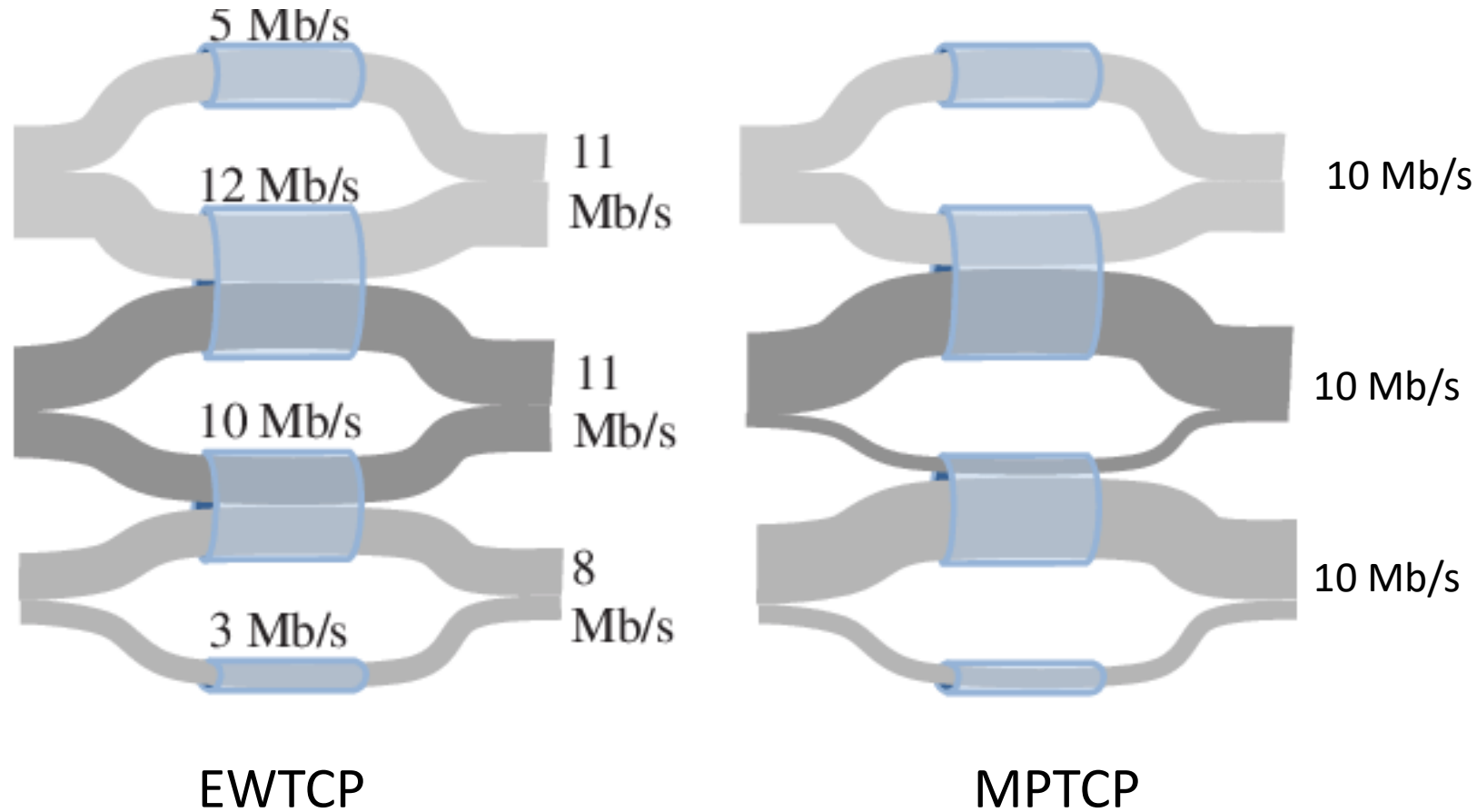
MPTCP congestion control

- Regular TCP
 - Each round trip, increase window w by 1
 - Each loss, decrease window w by $w/2$
- Multipath TCP
 - Each round trip, increase w_r by w_r/w_{total}
 - Each loss, decrease window w_r by $w_{total}/2$

MPTCP congestion control

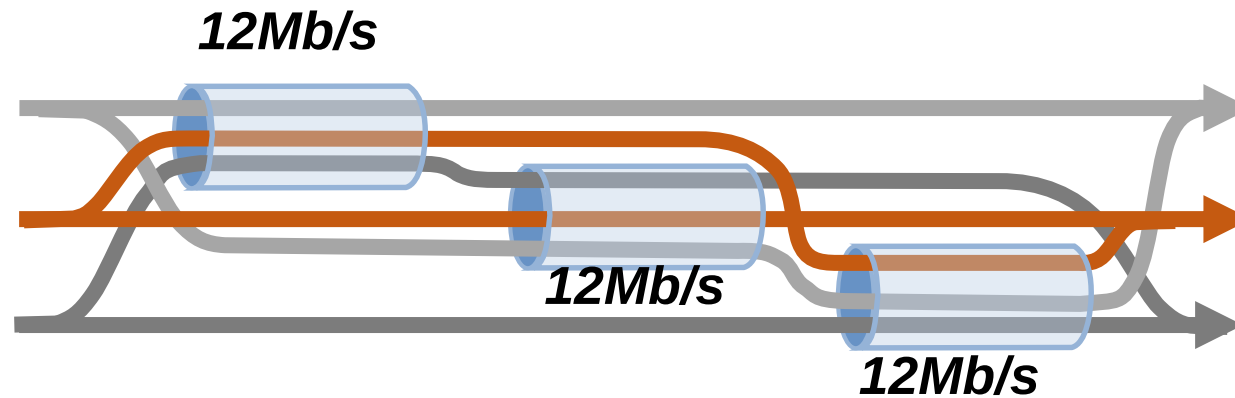
- “Resource pools,” links that act together as if they are a single, larger-capacity link shared by all flows.
- Allow less-congested subflows to increase proportionally more than the congested ones: resource pooling in the increase phase of the congestion control
- Total increase of multipath TCP across all of its subflows is dynamically chosen so that goal 1 and 2 are achieved

Comparison between EWTCP and MPTCP



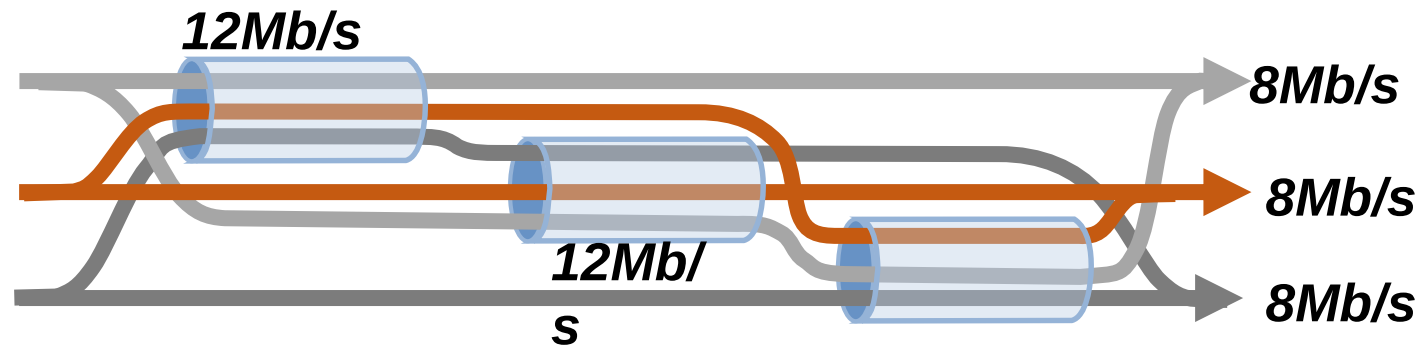
Img src: Design, implementation and evaluation of congestion control for multipath TCP Damon Wischik et al.

Use Efficient Paths



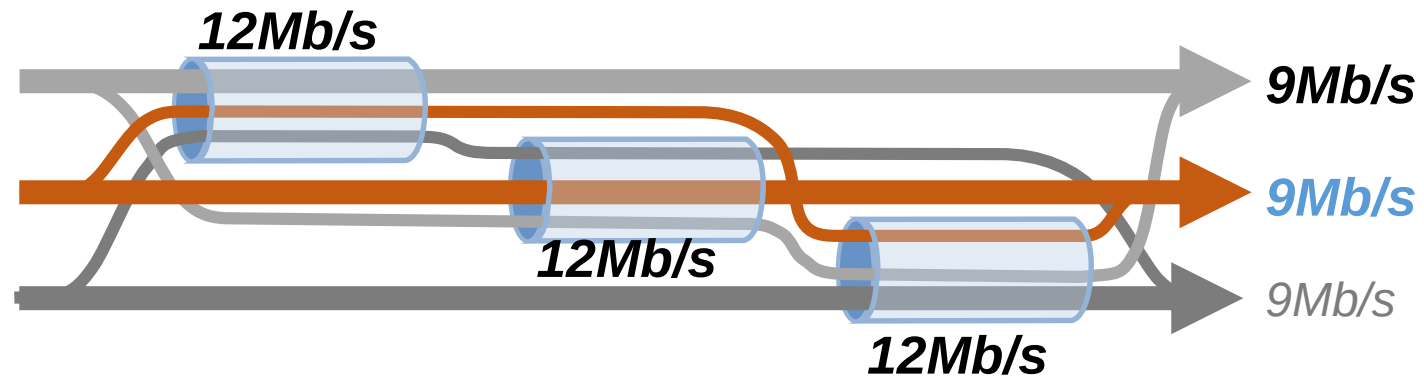
*Each flow has a choice of a 1-hop and a 2-hop path.
How should split its traffic?*

Use Efficient Paths



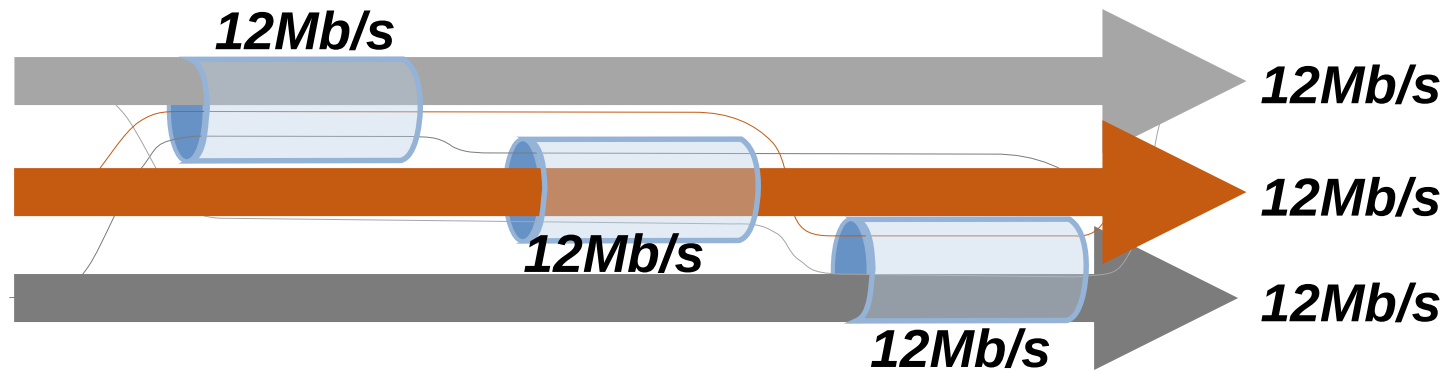
If each flow split its traffic 1:1 ...

Use Efficient Paths



Ack: Damon Wischik

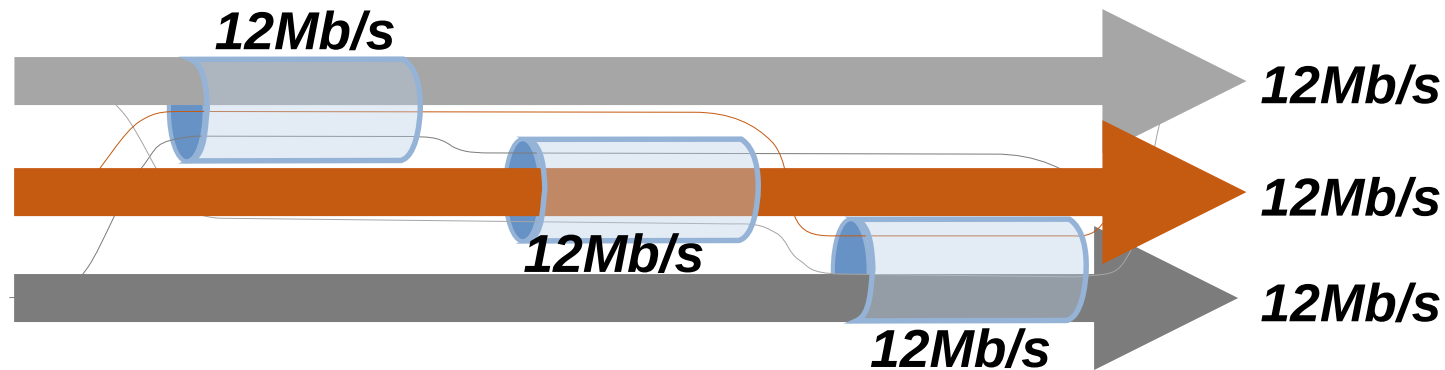
Use Efficient Paths



Better: Each connection on a one-hop path

Each connection should send all traffic on the least-congested paths

Use Efficient Paths



Better: Each connection on a one-hop path

Each connection should send all traffic on the least-congested paths

But keep some traffic on the alternate paths as a probe

MPTCP should send all its traffic on its least-congested paths